HastaDAQ strain-gauge board

Net labels connect everything: same name = same net. Companion to the main board (kicad/hastadaq). Reusable: any I2C master with 5 V and 3.3 V logic can drive it.

POWER: bulk + decoupling on the loom 5 V (any ferrite bead is part of the loom, not the board), the LDO makes the quiet 3.3 V that powers BOTH ADCs and the bridge excitation

J1
LOOM from main board (4-wire I2C): GND, +5V, SDA, SCL

GND 1
+5V 2
SDA 3
SCL 4
GND
5V
SDA
SCL

C1
47uF (>=10V)
+5V
GND

C2
100nF
+5V
GND

U1
3.3 V LDO: LP2950-N 3.3 V (TO-92: pin 1 OUT, 2 GND, 3 IN)
+5V 3
GND 2
IN
GND
OUT 1
+3V3

C3
10uF out
+3V3
GND

C4
100nF
+3V3
GND

#FLG1
PWR_FLAG
+5V

#FLG2
PWR_FLAG
GND

Pin 2 is +5V, NOT 3V3: use a keyed plug, never a Qwiic cable.

Load is about 20 mA (bridges 2 x 4.7 mA + ADCs): the LDO dissipates about 35 mW.

J2
LOOM OUT (pass-through, paralleled with J1): GND, +5V, SDA, SCL

GND 1
+5V 2
SDA 3
SCL 4
GND
5V
SDA
SCL

Second 1x4 header wired straight to the loom: daisy-chains another I2C board (+5V, 3.3 V logic).

U2
Adafruit ADS122C04 #1 (I2C 0x40)
+3V3 1
GND 2
SDA 3
SCL 4
5
6
VIN
GND
SDA
SCL
RST
DRDY
A0 7
A1 8
A2 9
A3 10
Ref+ 11
Ref- 12
ADS1_A0
ADS1_A1

ADS1 input filter (optional: 1k links + 100nF)
ADS1_SIG
ADS_REF
ADS1_A0

R1
0R link
ADS1_SIG
ADS1_A0

R2
0R link
ADS_REF
ADS1_A1

C5
DNP
ADS1_A0
ADS1_A1

ADS1 decoupling (at the ADC)

C6
10uF
+3V3
GND

C7
100nF
+3V3
GND

J3
HB1 half-bridge #1 (on beam): V, SIG, GND
+3V3 1
ADS1_SIG 2
GND 3
V
SIG
GND

Shared completion resistors (fixed) -> A1 of BOTH ADCs

R5
350R fixed
+3V3

R6
350R fixed
GND

ADS_REF

U3
Adafruit ADS122C04 #2 (address jumper bridged)
+3V3 1
GND 2
SDA 3
SCL 4
5
6
VIN
GND
SDA
SCL
RST
DRDY
A0 7
A1 8
A2 9
A3 10
Ref+ 11
Ref- 12
ADS2_A0
ADS2_A1

ADS2 input filter (optional: 1k links + 100nF)
ADS2_SIG
ADS_REF
ADS2_A0

R3
0R link
ADS2_SIG
ADS2_A0

R4
0R link
ADS_REF
ADS2_A1

C8
DNP
ADS2_A0
ADS2_A1

ADS2 decoupling (at the ADC)

C9
10uF
+3V3
GND

C10
100nF
+3V3
GND

J4
HB2 half-bridge #2 (on beam): V, SIG, GND
+3V3 1
ADS2_SIG 2
GND 3
V
SIG
GND

NOTES
1. Each half-bridge head (2 gauges) plugs into a 3-pin header: V (+3V3), SIG (mid-point), GND. The two fixed 350R completion resistors are shared; their mid-point (ADS_REF) goes to A1 of both ADCs. Excitation is the ADC's own VIN, so VREF_SUPPLY is ratiometric: no Ref+/Ref- wiring.
2. 3V3 comes ONLY from the on-board LDO (from the loom's 5 V): cable voltage drop and the main board's switching noise do not reach the bridges or the ADC reference. The ADC logic lines (SDA/SCL) are pulled up by the ADS122C04 breakouts to this 3V3 and are 3.3 V logic: the master must not drive them above 3.6 V.
3. Input filters are optional: with the 0R links and no capacitor fitted there is none (original behaviour). Fit 1k in each link and 100nF across A0-A1 for about an 800 Hz low-pass.
4. ADS #2: bridge one address jumper on its back (resulting address unconfirmed; the main board's firmware scans at boot).
5. Use shielded twisted cable for the gauge wires, shield to GND at this board only.

HastaArduino		
Sheet: /		
File: gaugeboard.kicad_sch		
Title: HastaDAQ strain-gauge board		
Size: A3	Date:	**Rev: A**
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